Time Series Decomposition

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DATA 624

Adjustments and Transformations

When decomposing a time series, it can be helpful to first transform or adjust the series to make the decomposition as simple as possible.

Calendar Adjustments

A bit of variation in seasonal data can be due to calendar effects, and this can be usually resolved by removing variation before further analysis.

- Example: A manufacturing plant wants to analyze season patterns, but total monthly productions are distorted due to differing numbers of operating days.

Population Adjustments

Data affected by population changes can be adjusted to give per-capita data. (Consider data per person, per thousand people, or per million people)

- Example: You are studying the number of beds per thousand people.

Inflation Adjustments

The data that is affected by value of money is best adjusted before modelling.

- Example: The average cost of a new house increased over a few decades because of inflation. A house that cost $100,000 in 2000, will cost more 30 years later.
- Financial time series are typically adjusted that values are adjusted in dollar values for a certain year.
  - $X_t = y_t / z_t * z_{2000}$
  - where t = year, z_t denotes price index and y_t denotes original house

Mathematical Transformations

For data showing variation that increases or decreases with the level of the series, a transformation can be useful.

- Denote original transformation as y_1, ..., y_t and the transformed observation s w_1, ..., w_t then w_t = log(y_t)

Time Series Components

Seasonal Component S_t

- Data following a pattern with fixed frequency

Trend-cycle component T_t

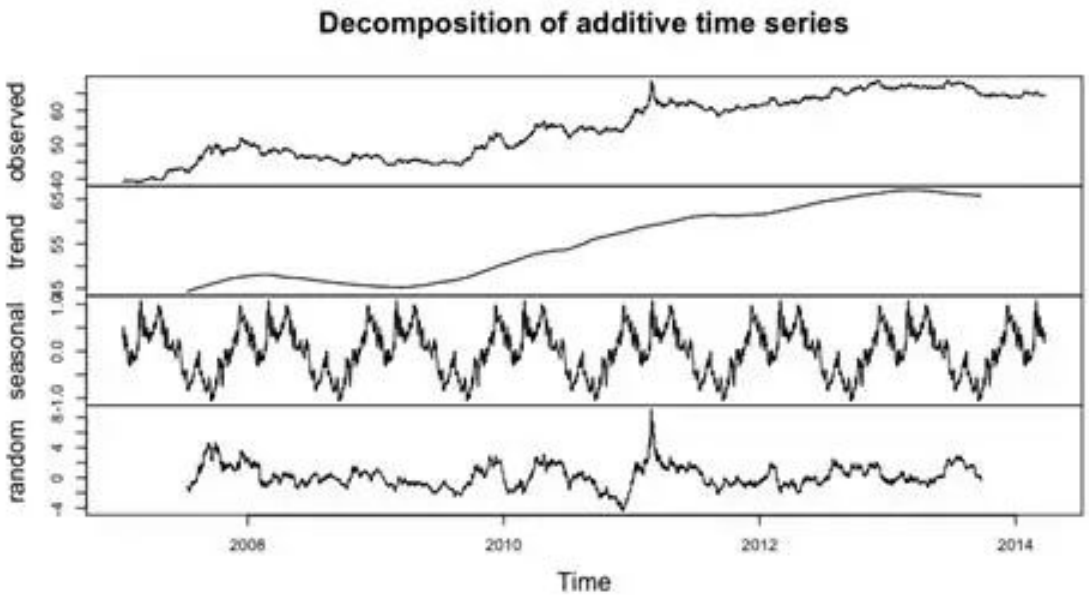
- Data following a increasing or decreasing pattern

Remainder Component R_t

- Variations impacting individual variables

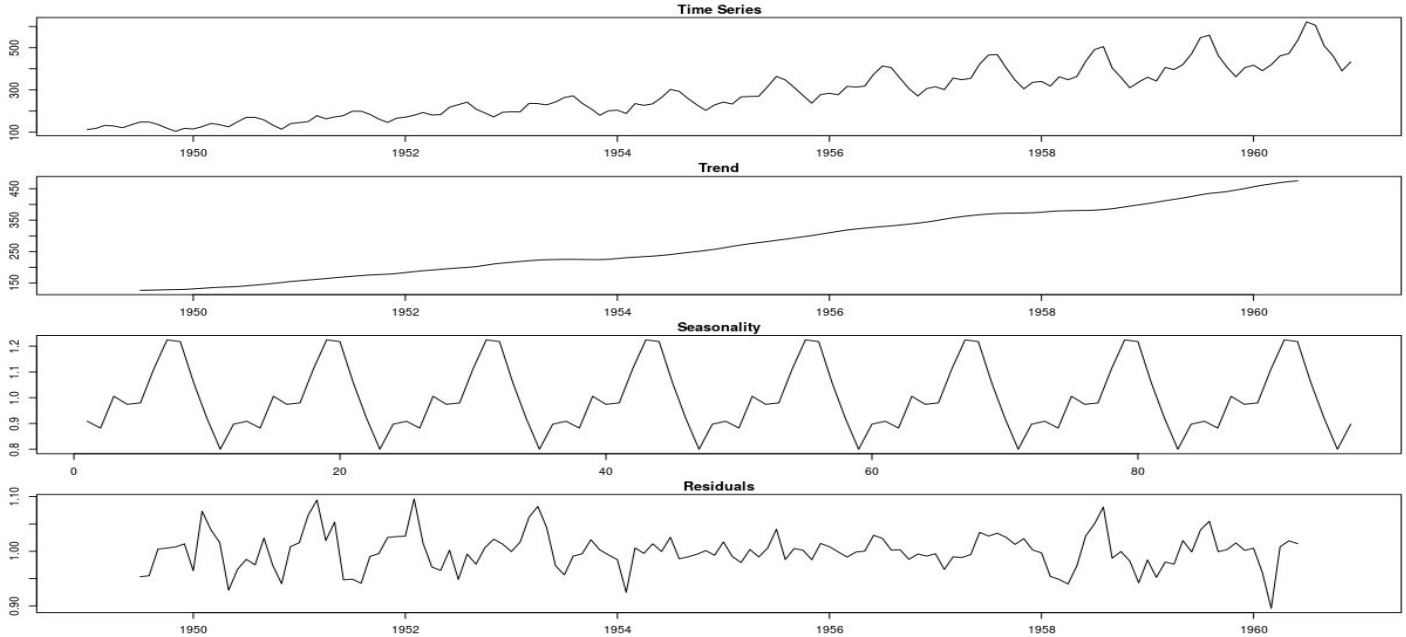
Additive Decomposition

$$Y_t = S_t + T_t + R_t$$



Multiplicative Decomposition

$$S_t * T_t * R_t = \log S_t + \log T_t + \log R_t$$



Moving averages: smoothing a trend-cycle

- **Definition:** An *m*-MA is the average of a time series within *k* periods of a center point.
- **The Formula:**

$$\hat{T}_t = \frac{1}{m} \sum_{j=-k}^{k} y_{t+j},$$

- **Purpose:** To "smooth" the data. By averaging nearby observations, we cancel out random fluctuations (noise) to see the underlying trend.
- **The "Trade-off":** Higher order (*m*) creates smoother trend, but causes loss of more data at the beginning and end of the series.

Moving averages: smoothing a trend-cycle

Pros

- **Extreme Simplicity:** Very easy to calculate in a spreadsheet or basic SQL; requires zero statistical parameters other than the window size.
- **Uniform Smoothing:** It provides a consistent "lag" that is easy to understand—a 10-day SMA is exactly the average of the last 10 days.
- **Effective for High Noise:** If data is extremely wiggly with no clear direction, an SMA is great at flattening that noise completely.

Cons

- **The Outliers Effect:** Because every point has equal weight, an outlier entering the window shifts the average, and then shifts it again when it leaves the window *m* days later, creating a "ghost" jump.
- **High Lag:** SMAs are slow to respond to rapid changes in trend because the old data (at the edge of the window) carries as much weight as today's data.

Centered Moving Averages (2 × *m*-MA)

- **The Symmetry Problem:** Even-order averages (like a 4-MA) don't have a natural center; they fall *between* periods.
- **The Solution:** Apply a second moving average (usually order 2) to "re-center" the data.
- **The Weighted Result:** A **2 × *5*-MA** becomes a weighted 5-point average.
- **The weights sum** must equal to 1.
- **Why it Works:** In quarterly data, the first and last terms represent the same quarter in different years. Giving them "half-weight" ensures no single quarter overpowers the average.

$$\hat{T}_t = \frac{1}{8}y_{t-2} + \frac{1}{4}y_{t-1} + \frac{1}{4}y_t + \frac{1}{4}y_{t+1} + \frac{1}{8}y_{t+2}$$

$$w = [\frac{1}{8}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{8}]$$

Centered Moving Averages (2 × *m*-MA)

Pros

- **Reduced Lag:** By placing more weight on the most recent (or central) observations, the trend line tracks the actual data more closely.
- **Superior Smoothing:** WMAs (like the Spencer's 15-point or Henderson moving averages) result in much smoother curves that don't "wiggle" as much as simple ones when data enters the window.
- **Mathematical Neutralization:** In 2 × *m*-MA, weighting is the only way to perfectly balance quarterly or monthly seasonality so that no single period overpowers the trend.
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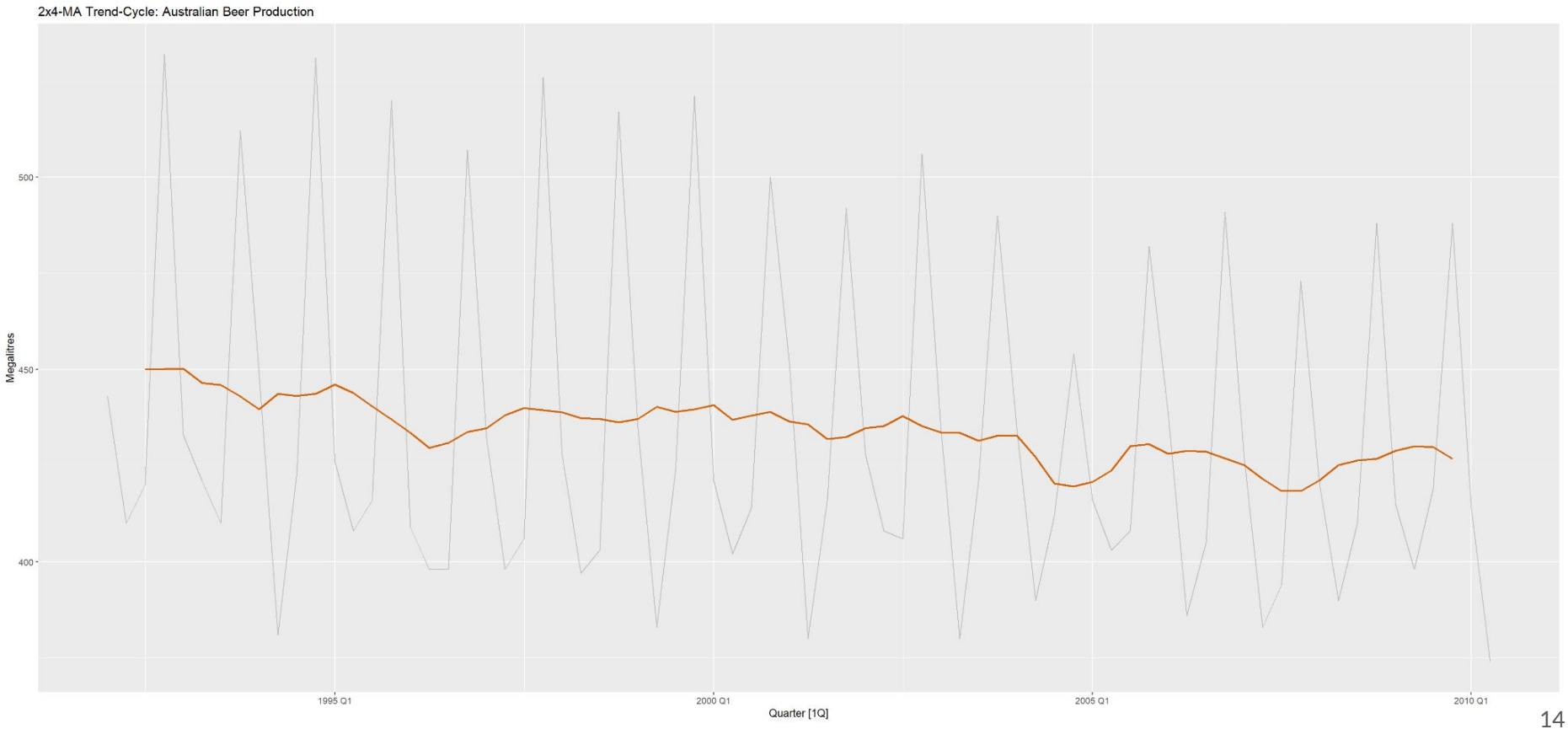
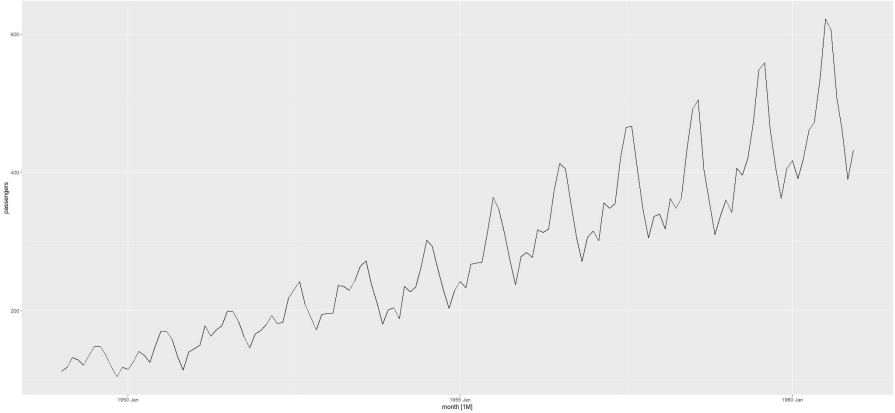
Cons

- **Complexity:** Harder to explain to a non-technical audience (e.g., "Why is this quarter worth twice as much as that one?").
- **Weight Selection:** Choosing the "correct" weights can be arbitrary unless you follow a specific standard (like classical decomposition).
- **Edge Data Sensitivity:** If the weights are heavily front-loaded, a single outlier can cause a massive, misleading spike in the trend line.

Centered Moving Averages (2 × *m*-MA)



ACF



ACF (raw series)

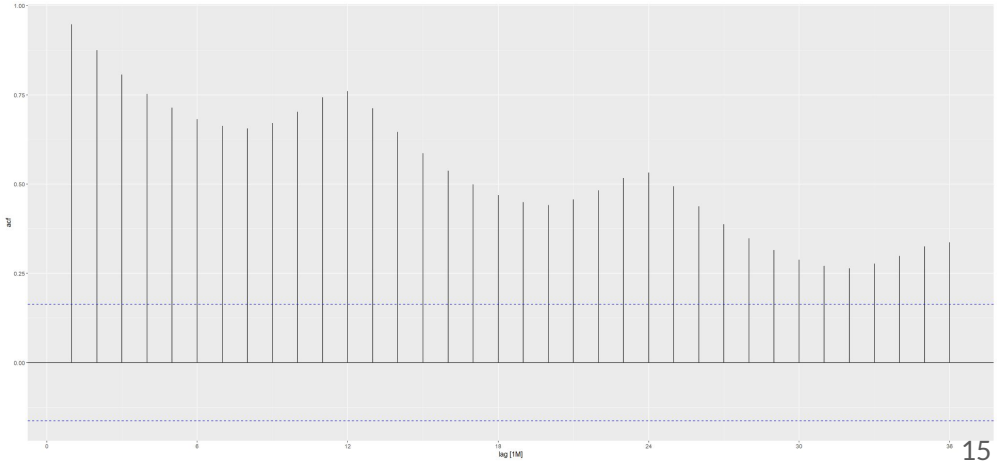
- Large positive autocorrelations at small lags
- Slow decay across many lags
- Indicates strong **trend** and **seasonal dependence**

Interpretation

- Trend causes correlations to persist across many lags
- Seasonality creates repeated dependence at fixed intervals
- Noise and structure are mixed together

Implication

We cannot diagnose the true dependence structure while components are overlapping.



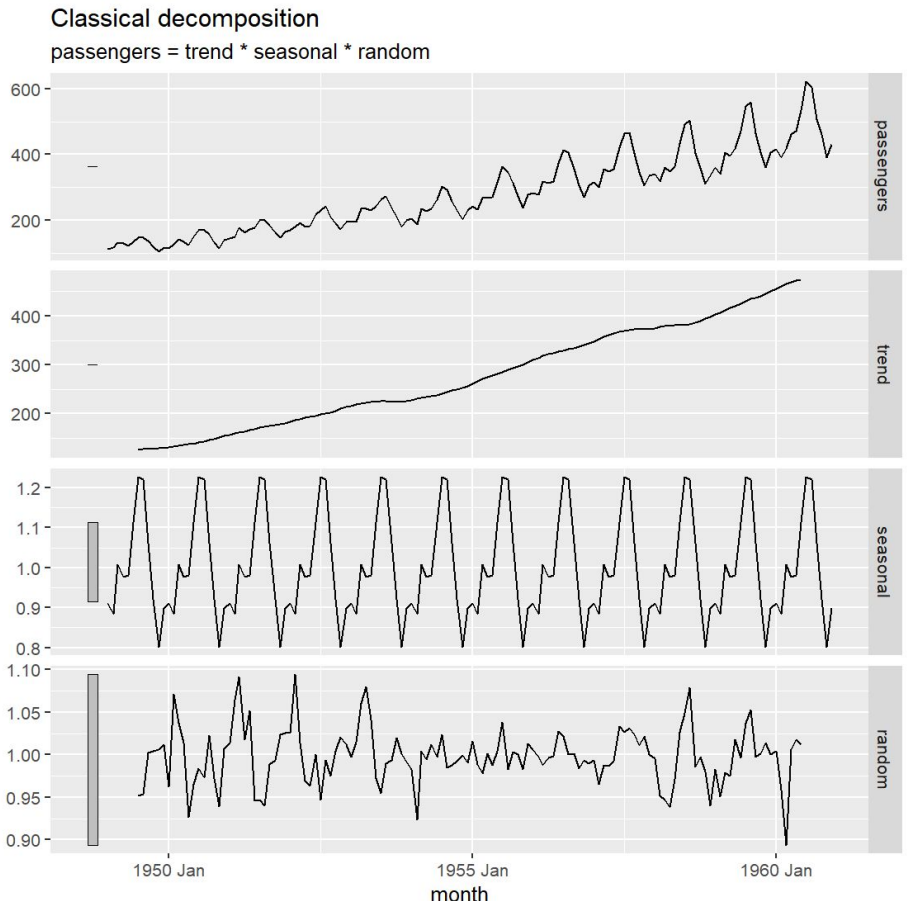
Classical Decomposition

What decomposition does

- Moving averages remove trend
- Seasonal averaging removes seasonal effects
- Leaves a remainder component

ACF after decomposition

- Trend in ACF is isolated in trend component
- Seasonal spikes are isolated in seasonal component
- Remainder ACF preferably ideally near zero



Pros and cons of Classical Decomposition

Classical Decomposition — Pros

- Simple and easy to explain
- Clearly separates trend, seasonality, and noise
- Helps visualize structure in the data
- Makes ACF patterns easier to interpret
- Useful as an exploratory tool before modeling
- Works well when seasonality is stable over time

Classical Decomposition — Cons

- Assumes seasonality does not change over time
- Trend is forced to be smooth, even if reality isn't
- Edge values are lost due to moving averages
- Sensitive to outliers
- Not a forecasting model by itself
- Can give a false sense of structure if data is mostly noise

Methodologies Used by Official Statistical Agencies

Evolution of Time Series Decomposition

- **1920s:** Development Time Series using **Classical Decomposition** methods.
- **1950s:** The **US Census Bureau** created the more robust "Census II" approach.
- This formed the foundation for the famous **X-11** method and its successors (**X-12-ARIMA** and **X-13-ARIMA**).
- **1990s:** The **TRAMO/SEATS** method was developed in Europe, offering a model-based alternative for seasonal adjustment.
- **STL Method: STL** (Seasonal and Trend decomposition using Loess) was designed by Bell Labs.

Methodologies Used by Official Statistical Agencies

X-11 Decomposition: Key Features

- **Algorithmic Iteration:** Uses a multi-step process rather than a single formula to refine components.
- **Advanced Moving Averages:** Employs sophisticated filters for smoother trend-cycle and seasonal isolation.
- **Evolving Seasonality:** Allows seasonal patterns to change over time, unlike rigid classical methods.
- **Outlier Robustness:** Automatically identifies and adjusts for extreme values and trading day effects.
- **Flexible Framework:** Handles both additive and multiplicative data; easily implemented via the X11() function in fable.

Methodologies Used by Official Statistical Agencies

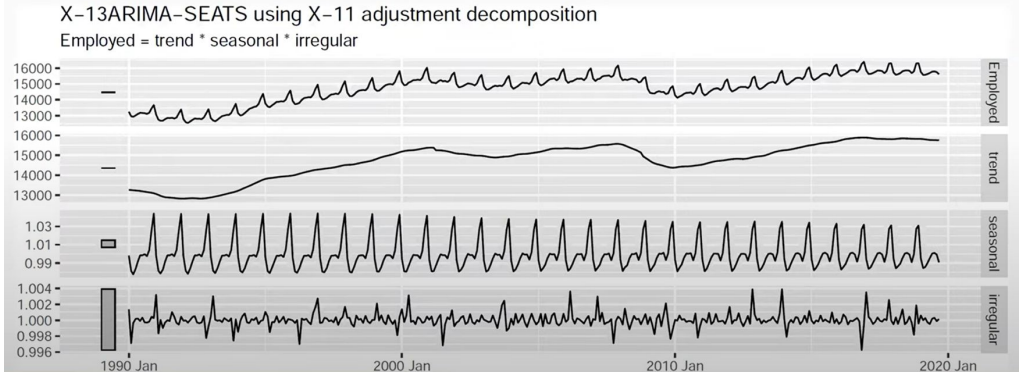
Disadvantages of X-11

- **No Formal Statistical Model:** Based on an empirical algorithm rather than a mathematical model.
- **No Prediction Intervals:** Does not provide measures of uncertainty for the components.
- **Restricted Frequency:** Limited to monthly and quarterly data only.
- **No Support for Covariates:** Cannot incorporate external variables (e.g., weather or strikes).
- **Black Box Nature:** The iterative steps can be difficult to trace mathematically compared to simpler methods.

Methodologies Used by Official Statistical Agencies

Decomposition of Total US Retail Employment using X-11

```
x11_dcmp <- us_retail_employment |>
  model(x11 = X_13ARIMA_SEATS(Employed ~ x11())) |>
  components()
autoplot(x11_dcmp)
```



Methodologies Used by Official Statistical Agencies

Advantages of SEATS Decomposition

- **Model-Based Approach:** Built on a formal statistical foundation (ARIMA models).
- **Statistical Rigor:** Provides theoretically sound components with calculated uncertainties.
- **Prediction Intervals:** Allows for the calculation of intervals for the trend and seasonal components.
- **Sophisticated Filtering:** Uses the specific characteristics of the data to determine the best filters.
- **Modern Standard:** Works seamlessly within the X-13ARIMA-SEATS framework used by global agencies.

Methodologies Used by Official Statistical Agencies

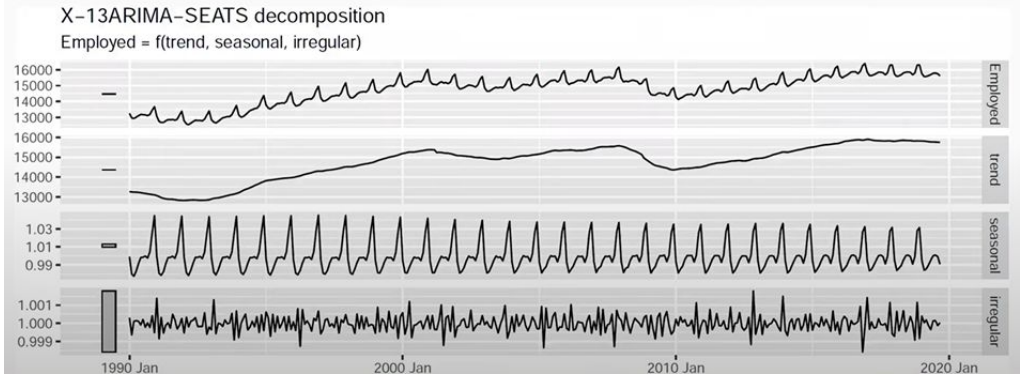
Disadvantages of SEATS

- **Complexity:** Requires advanced knowledge of ARIMA models to interpret.
- **Rigid Frequency:** Limited strictly to **monthly and quarterly** data.
- **Single Seasonality:** Cannot handle multiple seasonal patterns (e.g., daily + weekly).
- **Model Dependent:** Decomposition quality relies entirely on the initial ARIMA model fit.
- **"Black Box" Feel:** Harder to explain to non-technical audiences than simple averages.

Methodologies Used by Official Statistical Agencies

Decomposition of Total US Retail Employment using SEATS

```
seats_dcmp <- us_retail_employment |>
  model(seats = X_13ARIMA_SEATS(Employed ~ seats())) |>
  components()
autoplot(seats_dcmp)
```



Seasonal and Trend Decomposition Using Loess (STL)

- Versatile decomposition method
  - Captures short- and long-term variations
  - Handles complex patterns

- Iteratively applies Loess
  - Loess = Locally-Weighted Scatterplot Smoothing
  - Nonparametric regression method that fits curves to localized subsets of data

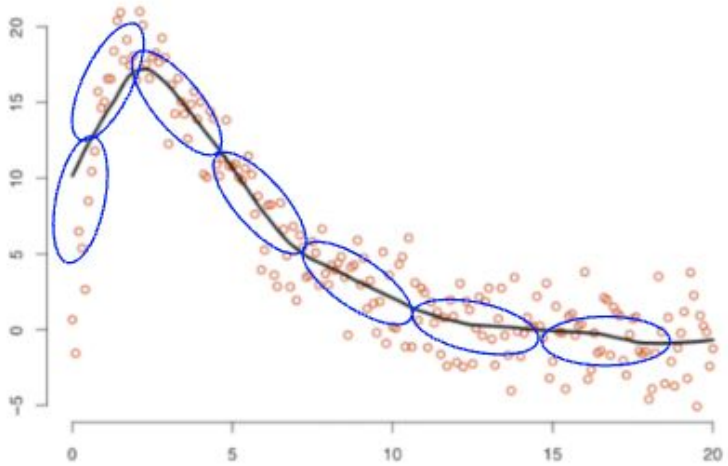


Image source: Lowess and Loess, Clearly Explained!!!
www.youtube.com/watch?v=Vf7oJ6z2LCc

Advantages and Disadvantages of STL

Advantages

- Handles any type of seasonality
- Highly flexible (eg, user can define seasonal component, smoothness)
- Robust to outliers

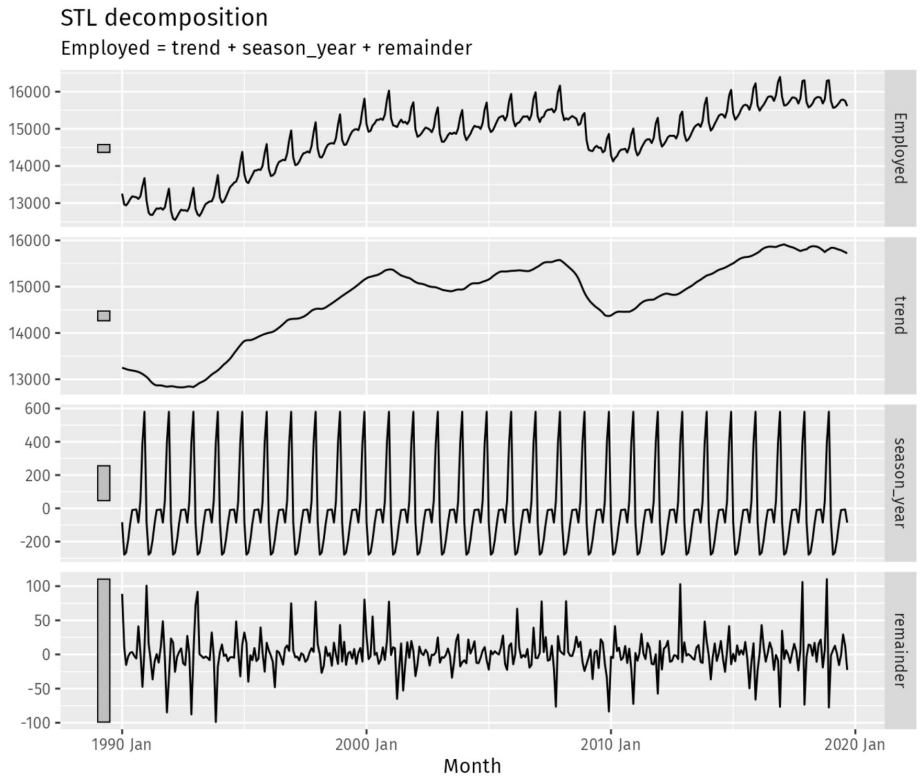
Disadvantages

- Does not handle trading day or calendar variations
- Mainly additive decomposition

STL Example

```
us_retail_employment |>
  model(STL(Employed ~
              trend(window = 7) +
              season(window = "periodic"),
        robust = TRUE)) |>
  components() |>
  autoplot()
```

- Trend and season window sizes should be odd numbers
- robust=TRUE makes Loess smoothing less sensitive to outliers



STL Features

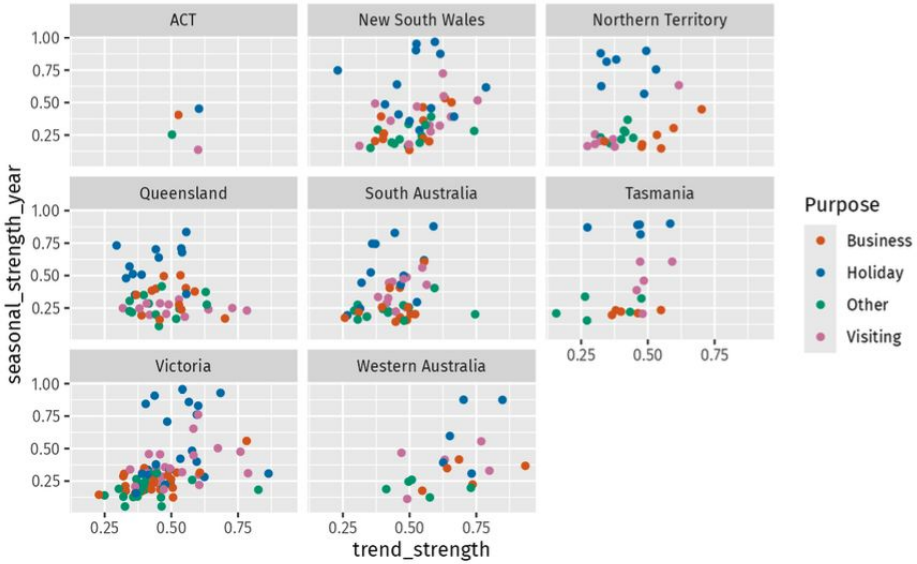
- Strength of trend

$$F_T = \max\left(0, 1 - \frac{\mathrm{Var}(R_t)}{\mathrm{Var}(T_t + R_t)}\right)$$

- Strength of seasonality

$$F_S = \max\left(0, 1 - \frac{\mathrm{Var}(R_t)}{\mathrm{Var}(S_t + R_t)}\right)$$

```
tourism |>
  features(Trips, feat_stl) |>
  ggplot(aes(x = trend_strength, y = seasonal_strength_year,
             col = Purpose)) +
  geom_point() +
  facet_wrap(vars(State))
```



R_t, remainder component; S_t, seasonal component; T_t, smoothed trend component; var, variance.

Source

Hyndman RJ & Athanasopoulos G. (2021). Forecasting: Principles and Practice, 3rd edition